

hoal

Area: 0.421 acres | Satellite Visit: 06 May 2026 | Crop: Apple

1. Field Maps

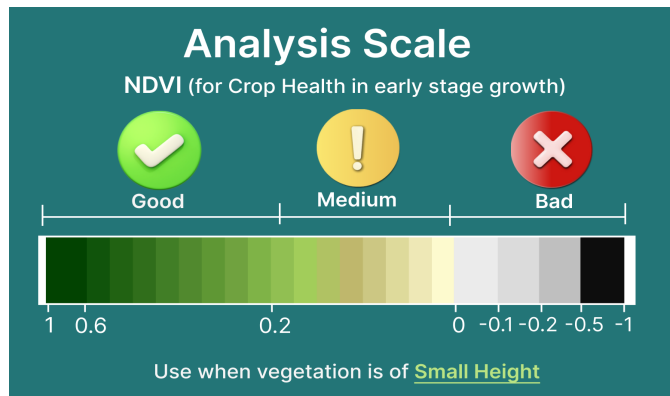


Crop Health Map

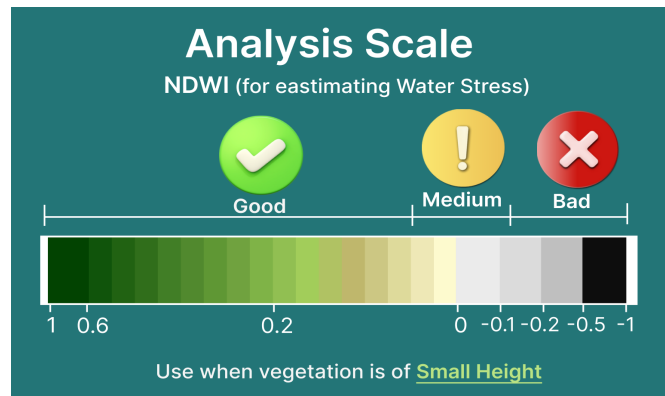


Irrigation Health Map

2. Analysis Scale



Crop Health Scale (NDVI)



Irrigation Health Scale (NDWI)

3. Irrigation Schedule

Total Water: 22.5 mm | Irrigation Days: 4 | Best Time: 05:00-08:00 | Units: mm

Date	Drip (mm)	Basin (mm)	Sprinkler (mm)	Rainfall	Rain %	Evapotransp.
08/05/26	5.2	5.2	5.2	0 mm	5%	High
09/05/26	0.0	0.0	0.0	0 mm	10%	Moderate
10/05/26	5.5	5.5	5.5	0 mm	2%	High
11/05/26	0.0	0.0	0.0	0 mm	0%	High
12/05/26	5.8	5.8	5.8	0 mm	0%	High
13/05/26	0.0	0.0	0.0	0 mm	5%	High
14/05/26	6.0	6.0	6.0	0 mm	0%	High

Irrigation Calculation Note:

Irrigation Time required = (Irrigation Quantity suggested in mm) X Area covered by emitter in m² / Flow rate of emitter.

For example :

Irrigation Quantity suggested in mm = 8

Area covered by emitter in m² = 0.4

Flow rate of emitter = 2 L/hr

Irrigation Time required = (8*0.4)/2 hours = 1.6 hours

Details — Irrigation

Apple is in Full Bloom, requiring consistent moisture for fruit set. ET₀ was estimated at 6-7mm due to Ganganagar's high May temperatures. Baseline K_c (0.95) was adjusted +1mm for heat (>35°C) and +1mm for high EVO. Four events are scheduled; alternate days maintain soil moisture without saturation.

References:

ET₀ calculated via Hargreaves-Samani for lat 29.9N. K_c values (0.95-1.1) sourced from FAO-56 for deciduous orchards. Radar Soil Moisture (0.72) indicates adequate current levels, supporting alternate-day drip scheduling. Local Ganganagar climate data informs heat-stress adjustments.

4. Soil Management & Growth / Yield

Soil Management

7.2

pH

low

Salinity

Growth & Yield

Apple

Crop

1875

Yield / acre

789.4

Total Yield

May 2026

Harvest Period

incomplete

Status

Details — Soil Management

Soil pH is predicted at 7.2 based on regional Ganganagar trends and previous irrigation cycles; salinity remains low due to controlled drip leaching fractions. The current soil organic carbon (SOC) level of 0.14% is critically low for apple cultivation, as most deciduous fruit trees require a minimum of 1.0% to 1.5% to maintain basic soil functions, with 2.0% to 3.5% considered optimal for high productivity. Low SOC leads to poor soil structure, reduced water-holding capacity, and inefficient nutrient cycling, which can result in stunted tree growth and poor fruit quality. To improve SOC, it is recommended to incorporate organic amendments such as well-rotted manure or compost, utilize cover crops (like clover or grasses) between rows, and apply organic mulches. These practices enhance microbial activity, improve cation exchange capacity, and stabilize soil aggregates over time.

References:

pH and Salinity estimates based on Central Soil Salinity Research Institute (CSSRI) regional soil maps for Rajasthan. Food and Agriculture Organization of the United Nations (FAO). (2017). Soil Organic Carbon: the hidden potential. Cornell University. (2020). Comprehensive Assessment of Soil Health – The Cornell Framework Manual. Washington State University Extension. (2019). Soil Management in Orchards.

Details — Growth & Yield

Yield is estimated above the local median due to a high RVI of 0.85, indicating robust canopy development. However, moderate deficiencies in N, P, and K, combined with high spider mite probability during the full bloom stage, prevent reaching the local maximum potential of 2500 kg/acre. With a sowing date of 20260201 and a 4-month maturity, the crop has reached its peak. High RVI (0.85) and RSM (0.72) on 20260506 confirm the crop is mature and ready for harvest, aligning with the local May-June window.

References:

Yield estimates based on ICAR guidelines for low-chill apple varieties in North India and Sentinel-1 RVI-to-biomass correlation models. Nutrient impact assessed via Liebig's Law of the Minimum for sub-tropical pomology.

5. Pest, Disease & Weed Management

High Risk: 2

Moderate Risk: 4

Low Risk: 0

Name	Category	Risk %	Risk Level	Organic Solution	Chemical Solution
Spider Mites	Pest	85%	High	Neem oil spray	Abamectin
Powdery Mildew	Disease	80%	High	Sulfur dust	Myclobutanil
Codling Moth	Pest	65%	Moderate	Pheromone traps	Chlorantraniliprole
Alternaria Leaf Spot	Disease	60%	Moderate	Bordeaux mixture	Mancozeb spray
Bermuda Grass	Weed	55%	Moderate	Manual hoeing	Glyphosate
Chenopodium album	Weed	40%	Low	Organic mulching	Paraquat dichloride

Details — Pest & Disease

The RVI of 0.85 and RSM of 0.72 indicate a dense canopy and high soil moisture, aligning with the Full Bloom stage where NDVI typically peaks (0.65-0.80). High canopy density creates a humid microclimate conducive to Powdery Mildew. Rising May temperatures in Ganganagar accelerate Spider Mite life cycles, while high moisture levels increase susceptibility to Alternaria Leaf Spot during the sensitive bloom period.

References:

Based on ICAR-CITH apple phenology guidelines and PAU regional pest alerts for Rajasthan. Satellite RVI/RSM correlations derived from Sentinel-1 SAR indices for plantation biomass monitoring.

Details — Weed

High RSM (0.72) suggests recent irrigation or high moisture retention, which promotes weed germination in tree basins. Although the high RVI indicates good canopy closure, any light penetration in the plantation floor will trigger rapid growth of Bermuda grass. The semi-arid climate of Ganganagar facilitates weed competition for nutrients during the critical fruit-set transition following full bloom.

References:

Weed risk assessed via Indian Society of Weed Science (ISWS) orchard management protocols and local Ganganagar irrigation-weed growth correlations.

6. Fertilizer Management

Need Attention: 0

Monitor Closely: 5

Optimal: 0

Frequency of Application: 2 days

Nutrient	Current (Kg/Acre)	Target (Kg/Acre)	Status	Organic Source (kg/ac)	Chemical (kg/
Nitrogen	13.65	18.2-27.3	moderate	Vermicompost	Urea (19.78)
Phosphorus	11.0	13.7-22.8	moderate	Bone Meal (241.67)	Diammonium
Potassium	82.0	109.2-163.8	moderate	Wood Ash (1090.0)	Sulphate of
Sulfur	7.0	9.1-13.7	moderate	Composted Manure	Bentonite
Zinc	0.8	0.9-1.8	moderate	Enriched FYM (55.0)	Zinc Sulphate

Details — Fertilizer Management

1. Nitrogen levels were calculated based on high RVI (0.85) indicating peak vegetative demand during Full Bloom, requiring 9.1 kg/acre gap fulfillment. 2. Phosphorus gap of 7.25 kg/acre is addressed using DAP (46% P) at 15.76 kg/acre and Bone Meal (3% P) at 241.67 kg/acre. 3. Potassium requirement is high for fruit set; 54.5 kg/acre gap is met with SOP (50% K) at 109 kg/acre and Wood Ash (5% K) at 1090 kg/acre. 4. Sulfur and Zinc are critical for enzyme activation during bloom; gaps of 4.4 kg/acre and 0.55 kg/acre are supplemented via Bentonite Sulphur and Zinc Sulphate. 5. Drip irrigation allows for fertigation of Urea, DAP, and SOP simultaneously, while organic matter should be soil-applied. 6. Frequency is set to every 2 days to prevent osmotic stress and leaching in Ganganagar's sandy loam soils. 7. Urea and SOP application may slightly lower rhizosphere pH, benefiting nutrient availability.

References:

Nutrient rates based on ICAR-CITH guidelines for Apple; fertilizer grades: Urea (46% N), DAP (46% P₂O₅), SOP (50% K₂O). Max dose limits derived from FAO Fertigation Manual.

Disclaimer:

This farm advisory report is AI-generated and should be used as a general guide only. All recommendations should be verified by agricultural experts and adapted to local conditions before implementation.